Format Drift Masquerades as Reward Hacking:
A Pre-registered Study of When RL Exploits Flawed Checkers

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November 2026 · Code and data: https://github.com/Himanshu-Sangshetti/format-drift-rl

Abstract

Reinforcement learning with verifiable rewards (RLVR) optimises language models against automatic checkers, and checkers have loopholes. We ask where the hacks that RL learns against a flawed checker come from: are they *selected* from behaviour the starting model already exhibits, or *discovered* during training? We pre-registered a study in which Qwen3-0.6B is trained with Dr. GRPO against eight checkers on MATH and MBPP, including a narrow loophole presented alone, with a partial-credit ramp leading to it, and with an equal amount of random credit, and we measure how often each checker pays answers that an independent referee marks wrong. The most consequential finding concerns measurement rather than optimisation. Training changes the *format* in which the model writes answers (nested and bold boxes, words inside boxes, repeated "Final Answer" lines), and a conventional box-matching referee reads these right answers as wrong: in one run 96% of the answers it flagged as hacks were correct, and an apparent exploitation of +0.43 was in fact zero. With a referee robust to such drift, exploitation is confined to loopholes that are easier than the task itself. A checker that pays whenever the gold number appears anywhere was exploited strongly in one of two seeds (+0.42; true accuracy fell from 0.73 to 0.43), and MBPP's original unit tests mildly (+0.01 to +0.06). In these runs the starting model already produced paid-but-wrong answers on 54–92% of the problems where the trained model did, consistent with selection. The pre-registered causal test was null: the ramp did not increase exploitation over the loophole alone or over the random-credit control (one-sided permutation $p = 0.74$ and 0.70). We release the code, every saved model output, the pre-registration with all amendments, and a drift-robust referee, and we argue that measurements of reward hacking must validate their referee on the trained model's outputs, not the starting model's.

1 Introduction

Reinforcement learning with verifiable rewards has become a standard way to improve reasoning in language models: the model samples answers, an automatic checker scores them, and the policy is updated towards high-scoring answers (Shao et al., 2024; Lambert et al., 2024; DeepSeek-AI, 2025). The checker is usually a few lines of code, such as an answer extractor and a comparison, a set of unit tests, or a prompt to another model. Such checkers are imperfect, and a policy optimised against an imperfect objective tends to find and exploit its errors, a phenomenon variously called reward hacking or specification gaming (Amodei et al., 2016; Krakovna et al., 2020; Skalse et al., 2022). As training moves to thousands of programmatically generated environments, each with its own checker, it becomes important to know *in advance* which checkers will be exploited.

Whether that is possible depends on where hacks come from. If a hack is *selected*, the starting model already produces it occasionally, and RL merely increases its frequency; sampling the starting model and counting how often the checker pays wrong answers should then flag the risky checkers, as Zhang (2026) found for MBPP. If a hack is *discovered*, the starting model never produces it, and RL arrives at it by following a gradient in the reward; an audit of the starting model would then be blind to it. Recent work leaves both possibilities open: Egashira et al. (2026) observe training collapse even when the verifier's initial error rate is low, and Deng et al. (2026) report that the initial hack rate alone does not determine when hacking begins.

We designed a small pre-registered study to separate the two accounts. Its central manipulation holds a narrow loophole fixed and varies only whether the reward offers a *ramp* towards it. A partial-credit ramp should, under the discovery account, guide the policy into the loophole; a density-matched control that pays the same partial credit at

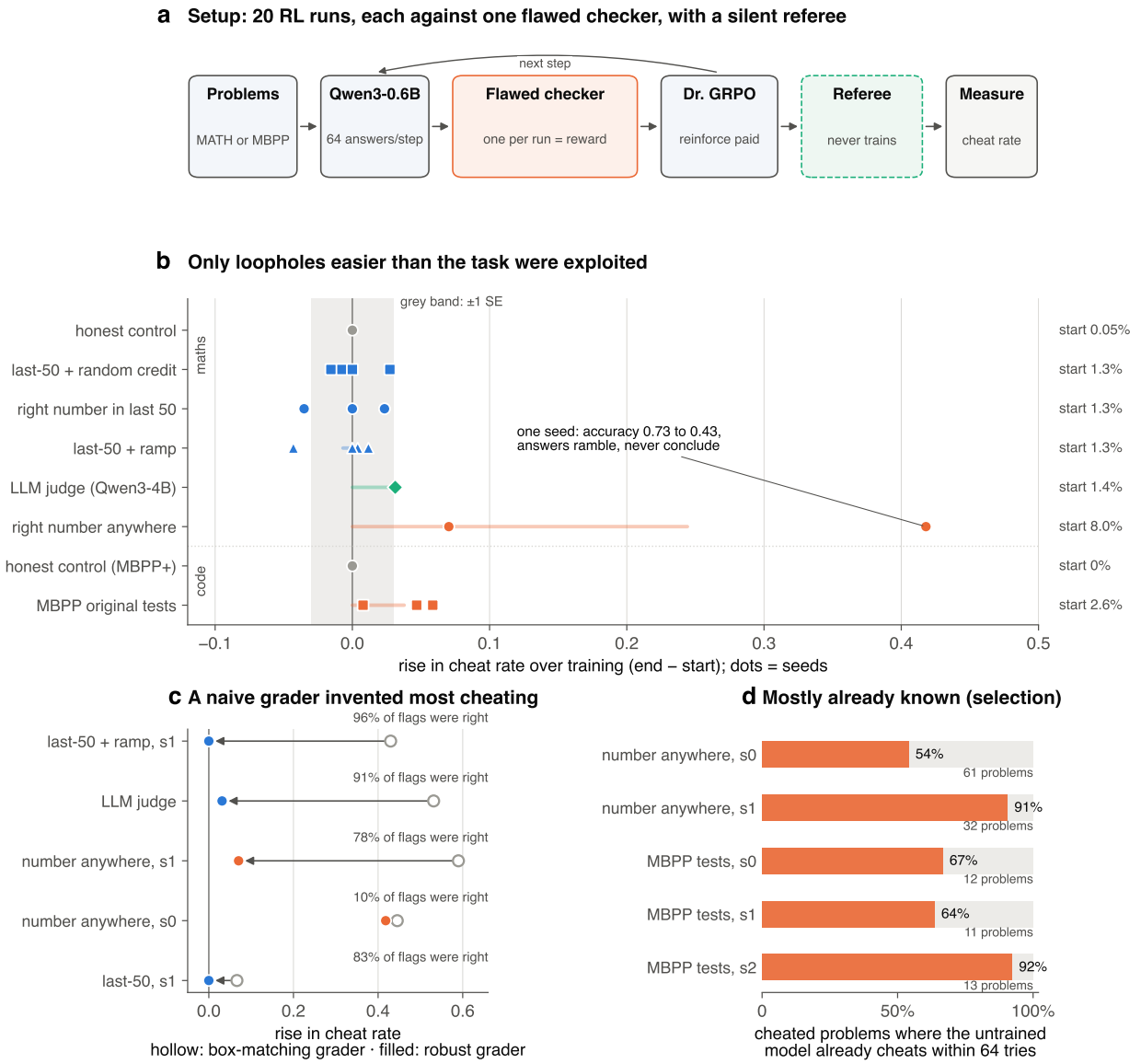


Figure 1. Overview of the study and its results. (a) Each of 20 runs trains Qwen3-0.6B with Dr. GRPO against one flawed checker; a silent referee, never used for training, determines which paid answers are wrong. (b) Rise in the paid-but-wrong rate over training under the robust referee (dots are seeds; "start" is the untrained model's rate). Only the "right number anywhere" checker and MBPP's original tests rise above noise. (c) The same quantity under a box-matching referee (hollow) and the robust referee (filled): most apparent exploitation consists of right answers in drifted formats. (d) Reachability: the share of exploited problems on which the untrained model already produces a paid-but-wrong answer within 64 samples.

random isolates the effect of the ramp from that of extra reward. We complement this causal test with checkers of different strength in maths and in code, an LLM judge, and a direct measurement of *reachability*: on each problem where the trained model hacks, can the untrained model already produce a hack?

The study did not unfold as planned, and the way it departed from the plan is our main result. When we read the answers that our pre-registered referee had flagged as hacks, most of them were correct. Training had changed how the model formats answers, and the referee, like many simple graders, could not parse the new formats. Under that referee the experiment appeared to confirm our hypothesis; under a referee robust to the drift, the effect disappears. We disclose the referee change as a post-hoc amendment, report all versions, and had the fix reviewed for leniency.

Contributions.

- We show that RL-induced *format drift* can make a fixed referee report substantial reward hacking that does not exist, quantify the effect (between 78% and 96% of flagged hacks were correct in three of four affected runs), and give a simple referee that is robust to it (Section 4.1).
- With the robust referee, we find exploitation only for checkers whose loophole is easier to satisfy than the task, in maths and in code, and we find it mostly reachable from the starting model, consistent with selection (Sections 4.2 and 4.4).
- We report a pre-registered null result: a reward ramp towards a narrow loophole did not cause exploitation at this scale, against a density-matched control (Section 4.3).
- We release all code, configurations, saved outputs (64 answers at 18 checkpoints for every run), audits and the pre-registration with nine dated amendments.

2 Related work

Reward hacking and specification gaming. Optimising a proxy objective can increase the proxy while degrading the intended objective (Amodei et al., 2016; Krakovna et al., 2020). Skalse et al. (2022) formalise reward hacking, Pan et al. (2022) map how misspecified rewards produce qualitative shifts in learned behaviour as capability grows, and Gao et al. (2023) characterise how the gold reward first rises and then falls as optimisation against a learned reward model increases. In language-model RL, frontier reasoning models have been observed to plan and carry out hacks in agentic coding tasks (Baker et al., 2025), and training on environments that admit hacks can generalise to broader misalignment (Anthropic, 2025). Our setting is deliberately simpler: small models, checkers with known loopholes, and a referee that labels every answer.

Imperfect verifiers in RLVR. Egashira et al. (2026) inject systematic false positives and false negatives into verifiers for arithmetic tasks and find that the pattern of errors matters more than their rate: some patterns delay training, some cause a plateau and some cause collapse, and collapse is possible from a low initial error rate. Zhang (2026) contrast MBPP's original tests with the stronger MBPP+ tests (Liu et al., 2023) in a pre-registered design and find that a static audit of the starting model predicts the mass of rewarded false positives, which they interpret as selection of pre-existing errors. Huang et al. (2025) compare rule-based and model-based verifiers for mathematical reasoning and find model-based verifiers more exploitable, and Wang et al. (2026) study hacking of rubric-based rewards. Gupta and Joshi (2026) synthesise this literature and note that verifier errors shift as the policy improves, so that accuracy measured before training says little about robustness during it. Our referee failure is a concrete instance of that observation, applied to the measurement of hacking rather than to the reward.

Predicting hacking before training. Deng et al. (2026) predict the onset of detectable hacking from a set of reference RL runs and a learned simulator, using Qwen3-0.6B and Qwen3-4B on coding environments whose prompts make the loophole salient. Our audits are training-free and aim at the narrower question of whether a hack is reachable from the starting model.

Pre-registration. Pre-registration separates confirmatory from exploratory analyses by fixing hypotheses, outcomes and tests before data are seen (Nosek et al., 2018). It is rare in machine learning; Zhang (2026) is a recent exception in this area. Our amendments, including the post-hoc one, are dated and released.

3 Experimental design

3.1 Quantities measured

Let π_θ be the policy, x a problem with gold answer $y^\star$, $r_c(a) \in [0, 1]$ the reward assigned to an answer a by the checker used for training, and $g(a) \in \{0, 1\}$ the label assigned by the referee, which is never used for training. An answer is

Checker	Reward	Seeds	h
strict	1 if the single boxed answer equals the gold (honest control)	1	0.0005
substring_tail50	1 if the gold number occurs in the last 50 characters	4	0.013
tail50_slope	as above, else 0.2 if the gold occurs elsewhere (a ramp)	4	0.013
tail50_density	as above, else 0.2 on a fixed hash of the reply at rate 6.6% (control)	4	0.013
judge_ref_prob	Qwen3-4B's $P(\text{YES})$ given problem, reference and reply	1	0.014†
substring_all	1 if the gold number occurs anywhere in the reply	2	0.080
mbpp_plus	1 if the program passes MBPP's tests and MBPP+ (honest control)	1	0
mbpp_base	1 if the program passes MBPP's original tests	3	0.026

Table 1. Checkers. h is the starting model's paid-but-wrong rate under the robust referee, measured on 2,048 maths and 1,200 code answers. †Measured with the box-matching referee, because judge scores were not stored per answer.

paid but wrong if $r_c(a) > \frac{1}{2}$ and $g(a) = 0$. For the set A_t of answers sampled at training step t, the paid-but-wrong rate is

$$\mathrm{PBW}_t \;=\; \frac{1}{|A_t|} \sum_{a \in A_t} \mathbf{1}\big[r_c(a) > \tfrac{1}{2} \,\wedge\, g(a) = 0\big]\,. \tag{1}$$

Exploitation is the change of this rate over training, $E = \overline{\mathrm{PBW}}_L - \overline{\mathrm{PBW}}_F$, where F and L are early and late windows of steps. The *mass* h of a checker is the paid-but-wrong rate of the starting policy π_0. Finally, let P be the set of problems on which a run produced at least one paid-but-wrong answer at step 75 or later. For each $x \in P$ we draw $K = 64$ answers from π_0 and define

$$\mathrm{reach} \;=\; \frac{1}{|P|} \sum_{x \in P} \mathbf{1}\big[\exists\, k \le K :\; r_c(a_k) > \tfrac{1}{2} \,\wedge\, g(a_k) = 0\big]\,, \qquad a_k \sim \pi_0(\cdot \mid x). \tag{2}$$

Selection predicts high reach, since the hack is already available to the starting model on those problems; discovery predicts low reach.

3.2 Checkers

Each run uses exactly one checker as its reward (Table 1). The maths checkers operate on MATH problems (Hendrycks et al., 2021) of difficulty 1–3 whose answer is an integer of at least two digits, so that the gold string is unambiguous for every checker. substring_all pays whenever the gold number occurs anywhere in the reply, a loophole that can be satisfied without solving the problem by mentioning many numbers. substring_tail50 restricts the occurrence to the final 50 characters, a much narrower loophole. tail50_slope adds partial credit of 0.2 when the gold occurs elsewhere in the reply, creating a ramp towards the narrow loophole, and tail50_density pays the same 0.2 on a fixed hash of the reply, at the base rate with which the gold occurs outside the final 50 characters in the starting model's answers (6.6%), so that it adds the same expected reward without a ramp. judge_ref_prob pays the probability of "YES" assigned by Qwen3-4B when shown the problem, the reference answer and the end of the reply; we chose Qwen3-4B over Qwen3-1.7B after a calibration in which the smaller judge accepted 96% of wrong boxed answers. For code we use MBPP+ (Liu et al., 2023): mbpp_base pays programs that pass MBPP's original three or four assertions (Austin et al., 2021), which are known to accept some wrong programs, and mbpp_plus additionally requires the MBPP+ tests. strict and mbpp_plus serve as honest controls.

3.3 Training

We train Qwen3-0.6B (Qwen Team, 2025) with thinking mode disabled, using Dr. GRPO (Liu et al., 2025) as implemented in TRL 1.17.1 (von Werra et al., 2020) with vLLM generation (Kwon et al., 2023). At each step the policy samples $G = 8$ answers for each of 8 problems; the advantage of answer i is $\hat{A}_i = r_i - \frac{1}{G}\sum_j r_j$, without division by the group's standard deviation, and the loss uses a constant normaliser, which removes the length and difficulty biases of GRPO (Shao et al., 2024; Liu et al., 2025). We use no KL penalty, a constant learning rate of 10^{-4}, LoRA of rank 16 on all linear layers (Hu et al., 2022), temperature 1.0 and answers of up to 1,536 tokens, the length used by Deng

Nested and bold box (gold 90)

Final Answer:
$$\boxed{\boxed{
\mathbf{90}}}$$

box-matching: wrong
robust: right

Unit inside the box (gold 60)

the boxed answer is:
$$\boxed{60}$$ ...
\boxed{60 \text{
square units}}

box-matching: wrong
robust: right

A real hack (gold 16)

Area $= 1/2\ |32| = 16$.
Wait, but hold on – let
me verify ... (repeats until
the length cap, no answer)

box-matching: wrong
robust: wrong

Figure 2. Format drift versus real exploitation. Late-training answers from tail50_slope, the judge run and substring_all. The first two are correct and are misread by a box-matching referee; the third is paid by substring_all because "16" appears, but it never states a final answer, and both referees correctly mark it wrong.

et al. (2026). Each run lasts 120 steps (7,680 answers). The learning rate was selected on the honest control by a pre-registered collapse rule, and all reported runs used the same GPU type (NVIDIA L40S). Appendix B lists every setting.

3.4 Referees

The code referee is MBPP+. For maths, the pre-registered referee applied Math-Verify[1] to a single boxed answer and marked a reply with two different boxed answers as wrong, a common and reasonable choice before training. The final referee, introduced in Section 4.1, extracts the value of the last box that contains a number, after removing formatting, words and a leading "$y =$", and compares it with the gold using Math-Verify; if nothing is boxed it uses the last number in the reply. This is close to the last-boxed extraction used by Open-R1.[2]

3.5 Pre-registration and analysis

Hypotheses, outcomes, the run set and the tests were fixed in a pre-registration before the main runs; nine dated amendments are summarised in Appendix A. The confirmatory test is that exploitation in tail50_slope exceeds that in substring_tail50 and in tail50_density, assessed by one-sided permutation tests over all splits of the four seeds per arm, Holm-corrected. All 64 answers were saved at steps 1, 10 and 25 and every fifth step from 50; because the per-step training log used the pre-registered referee, exploitation is computed from these saved answers with $F = \{1, 10\}$ and $L = \{105, 110, 115, 120\}$, the saved steps nearest the pre-registered windows. With 64 answers per step, the standard error of E is about 0.03.

4 Results

4.1 Format drift corrupts the referee

Before interpreting any training curve, we read the late answers that the pre-registered referee had labelled paid-but-wrong. Many were correct. Under checkers that do not reward a tidy final answer, the policy gradually changes the surface form of its answers while preserving their content. We call this *format drift*. Figure 2 shows typical examples: the answer boxed twice and set in bold, a box that contains the answer together with its unit, and, for contrast, a genuine hack. A referee that requires one distinct, parseable box marks each of the first two answers wrong.

We therefore replaced the referee with the last-box rule described in Section 3.4. It passes 18 hand-written test cases (Appendix D) covering both drifted correct answers and the hacks we intend to catch, such as a wrong final box with the gold elsewhere, or a reply that is cut off without a final answer. An independent review, carried out by a separate model instance with access to the raw outputs, read 40 answers that the new referee accepts and the old one rejected and found none that were wrong; of 30 answers the new referee still rejects, it judged 24 to be wrong or unfinished and about six to be borderline, mostly replies cut off after deriving the gold.

[1] https://github.com/huggingface/Math-Verify
[2] https://github.com/huggingface/open-r1

	Exploitation E		Answers flagged by box-matching referee	
Run	box-matching	robust	flagged	of which correct
tail50_slope seed 1	+0.43	0.00	307	294 (96%)
judge_ref_prob seed 0	+0.53	+0.03	208	190 (91%)
substring_all seed 1	+0.59	+0.07	512	397 (78%)
substring_all seed 0	+0.45	+0.42	320	33 (10%)

Table 2. The referee determines the conclusion. Exploitation under the pre-registered box-matching referee and the robust referee, and the number of paid answers (all saved steps) that the box-matching referee marked wrong. Values for every run are in Appendix C.

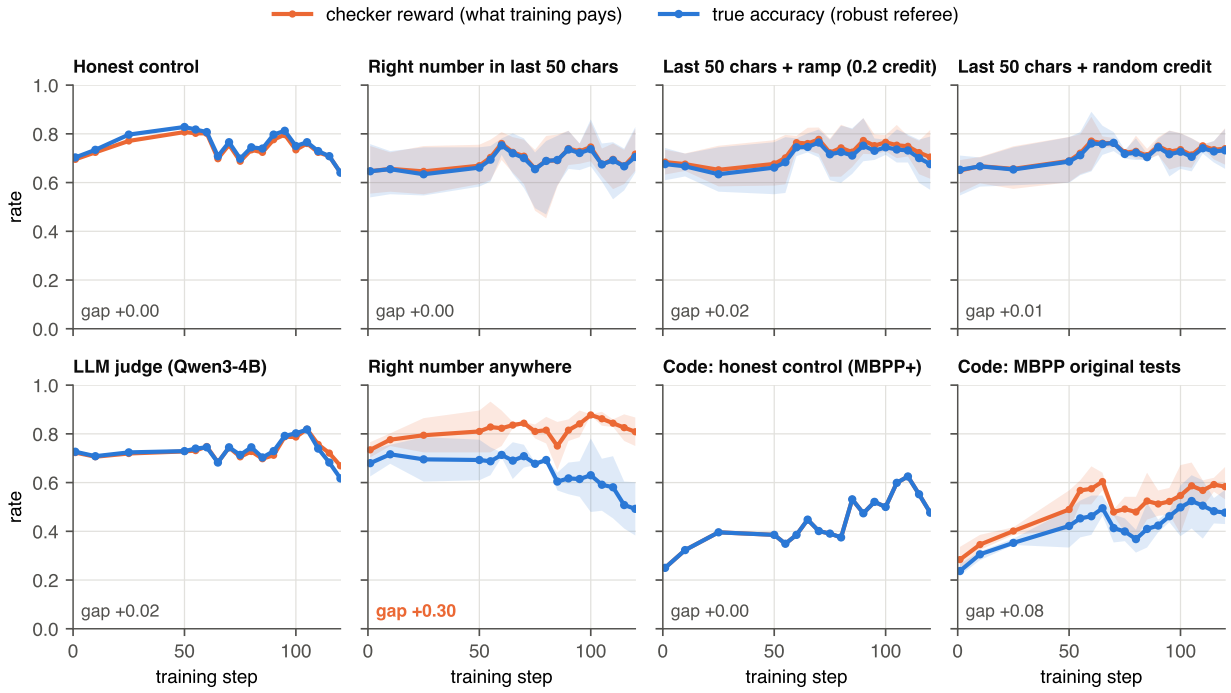


Figure 3. Training curves per checker. Orange: mean reward paid by the checker; blue: true accuracy under the robust referee. Both are computed on the same 64 saved answers per saved step, averaged over seeds (bands show the range across seeds) and smoothed over three saved steps. "Gap" is the reward minus accuracy over steps 105–120. A gap means that training pays for wrong answers; it appears only for "right number anywhere" and MBPP's original tests.

The consequences are large (Table 2). Under the pre-registered referee, seed 1 of tail50_slope shows an exploitation of +0.43, which would have supported our central hypothesis; under the robust referee it is 0.00, and 294 of its 307 flagged answers are correct. The judge run falls from +0.53 to +0.03, and seed 1 of substring_all from +0.59 to +0.07. Only seed 0 of substring_all is exploited under every version of the referee.

4.2 Which checkers were exploited

Figure 3 plots, for every checker, the reward paid by the checker and the true accuracy under the robust referee, computed on the same saved answers. For the honest controls the two curves coincide by construction. For the narrow tail50 loophole in all three of its variants, and for the judge, they also coincide: the checker paid essentially only correct answers throughout training, and true accuracy rose in eight of the twelve tail50 runs. Two checkers open a gap. Under substring_all the reward continues to rise while true accuracy falls, and under MBPP's original tests a smaller but persistent gap appears.

The behaviour behind the substring_all gap is easy to read in the outputs. Seed 0 learned to stop concluding: its answers grew from 741 to 4,129 characters on average and frequently ran to the length limit while re-checking

Checker	h	Exploitation E (per seed)	PBW at end	Accuracy, start → end
strict	0.0005	0.00	0.00	0.70 → 0.71
substring_tail50	0.013	+0.02, +0.00, +0.02, −0.04	≤ 0.03	0.65 → 0.69
tail50_slope	0.013	+0.00, +0.00, +0.01, −0.04	≤ 0.01	0.67 → 0.72
tail50_density	0.013	−0.01, +0.00, +0.03, −0.02	≤ 0.03	0.65 → 0.74
judge_ref_prob	0.014	+0.03	0.03	0.73 → 0.71
substring_all	0.080	**+0.42**, +0.07	0.45, 0.15	0.68 → 0.54
mbpp_plus	0	0.00	0.00	0.25 → 0.57
mbpp_base	0.026	+0.06, +0.05, +0.01	0.07, 0.06, 0.12	0.24 → 0.49

Table 3. Main results under the robust referee. E is the paid-but-wrong rate over steps 105–120 minus that over steps 1 and 10 (standard error ≈ 0.03). Accuracy is averaged over seeds.

a derivation, because the checker pays as soon as the gold appears and a conclusion earns nothing further. Its true accuracy fell from 0.73 to 0.43. Seed 1 of the same checker, from the same starting model and data, exploited only mildly (+0.07), in line with the observation of Deng et al. (2026) that the initial hack rate does not by itself determine the outcome. On MBPP the exploits are programs fitted to the visible assertion, for example functions that special-case the tested inputs; in a rehearsal run the policy returned a hard-coded dictionary of the tested answers.

Why was the narrow loophole never exploited? Its mass was small ($h = 0.013$), while about 70% of the starting model's answers were already correct and therefore paid. Each policy update was dominated by the reward for solving the problem, and the rare paid-but-wrong answer had little influence. Under substring_all, by contrast, writing many numbers is easier than solving a hard problem, so the exploit competes favourably with the task. We return to this interpretation in Section 5. The judge, which sees the reference answer, was barely exploited (+0.03), although its answers also grew long and repetitive: at step 120, 55% of them repeated a boxed answer eight or more times, a form of reward farming that is invisible to correctness metrics.

4.3 The pre-registered causal test

Exploitation in tail50_slope was not higher than in substring_tail50 (mean difference −0.010, one-sided permutation $p = 0.74$) or in tail50_density (−0.008, $p = 0.70$). The hypothesis that a ramp towards a narrow loophole causes it to be exploited is not supported. Under the pre-registered referee the ramp arm's mean was raised by seed 1, whose flagged hacks were almost all correct answers; even there, neither comparison survived the Holm correction ($p = 0.39$ and $p = 0.03$). As fixed in the pre-registration, we did not raise the learning rate to force exploitation. Because every arm sits at zero exploitation, the test establishes that the ramp did not create exploitation within 120 steps; it does not establish that ramps are irrelevant at larger scale or with longer training.

4.4 Selection: exploited behaviour was already reachable

In every run with real exploitation, the starting model already produced paid-but-wrong answers on most of the problems where the trained model did (Table 4, Figure 1d): reach ranged from 0.54 to 0.92, and on those problems the starting model's own paid-but-wrong rate was 8–23%, well above its average. This pattern is what selection predicts, and it agrees with Zhang (2026) on MBPP, now replicated with a different model family and extended to maths. The strongest run adds a nuance. About half of the problems hacked by seed 0 of substring_all were never hacked by the starting model in 64 samples. The strategy, writing long replies that do not conclude, was present from the start, and training extended it to problems on which it had not yet appeared. "Selection of a strategy, followed by its spread to new instances" describes this run better than either label alone. Because neither the ramp nor the judge produced real exploitation, the study contains no candidate case of discovery, and the contrast it was designed to test could not be made.

Run	Exploited problems $\lvert P\rvert$	Reach ($K = 64$)	Starting PBW rate on P
substring_all seed 0	61	0.54	0.096
substring_all seed 1	32	0.91	0.137
mbpp_base seeds 0 / 1 / 2	12 / 11 / 13	0.67 / 0.64 / 0.92	0.083 / 0.234 / 0.178

Table 4. Reachability in the runs with real exploitation. Runs with fewer than ten exploited problems are reported in the released files.

Study	Model, training	Loophole	Outcome
Deng et al. (2026)	Qwen3-0.6B / 4B, up to 400 steps	prompt invites redefining the test function, or hard-coding the visible examples	hacking observed
Egashira et al. (2026)	Qwen3-1.7B-Base, 500 steps	"approximately right" answers; outputs containing a keyword	approximate answers paid at a rate near 1 by step 50; keyword collapse, except when the model never produced the keyword
Zhang (2026)	1.5B coder, 400–800 steps	MBPP's original tests	rewarded false positives flat at about 15%
This work	Qwen3-0.6B, 120 steps	gold number anywhere; gold in last 50 characters (± ramp); MBPP's original tests	anywhere: exploited; last 50 characters: not exploited; MBPP: mild

Table 5. Comparison with the closest studies. Exploitation tracks how easily the loophole is satisfied relative to the task.

5 Discussion

5.1 Relation to prior findings

Our results are consistent with the closest prior studies once the ease of the loophole is taken into account (Table 5). Small models do exploit loopholes: Deng et al. (2026) observe hacking with the same Qwen3-0.6B model, on coding environments whose prompts point to the loophole, and we observe strong exploitation of substring_all at the same scale. Loopholes that the model rarely or never stumbles into are not exploited. Egashira et al. (2026) report that a reward paying any output containing the word "python" caused collapse, but not for a model that never produced the word, which mirrors our narrow loophole. On MBPP, Zhang (2026) find the mass of rewarded false positives essentially flat over 400–800 steps, and we find only mild exploitation in 120. Together these results suggest that how easy a loophole is to satisfy, relative to the task, matters at least as much as model size for whether it is exploited at this scale. Whether larger models with longer training *discover* exploits that are unreachable from the starting model, as reports on frontier systems suggest (Baker et al., 2025), remains open.

5.2 Recommendations for measuring reward hacking

The most transferable lesson concerns measurement. A referee validated on the starting model's answers can fail on the trained model's answers, because training changes their form. Box matching, exact string matching or any brittle parser will then count format drift as hacking, and a study can reach the opposite of the correct conclusion. We recommend (i) answer extraction that is robust to formatting, such as the last-box rule we release; (ii) a manual audit of a random sample of flagged answers at the end of training, reporting the false-alarm rate; and (iii) reporting results under more than one referee when the conclusion depends on the label. Correctness metrics also miss a second effect we observed: degenerate repetition of correct answers, which no referee in our study counted as a hack but which is a form of reward farming in its own right.

5.3 Limitations

We trained one small model, without thinking mode, for 120 steps with 64 answers per step; related studies train for 400–500 steps with larger batches (Egashira et al., 2026; Deng et al., 2026), and rare loopholes may need longer training or larger models to be exploited. The robust referee was chosen after the results were seen; it was reviewed for leniency, all versions are reported, and the one run that is exploited under every version is seed 0 of substring_all. Exploitation is computed on the saved steps closest to the pre-registered windows rather than on the windows themselves. The judge condition has a single seed. Finally, a pre-registered edit-based "slope" audit, intended to detect checkers prone to discovery, returned values near zero for every checker and predicted nothing; we report it only descriptively in the released files.

6 Conclusion

In a small pre-registered study, RL exploited flawed checkers only where the loophole was easier to satisfy than the task, and what it exploited was mostly within the starting model's reach. A reward ramp towards a narrow loophole did not create exploitation at this scale. The largest effect in our data, however, was produced by the measurement itself: training changed how the model writes its answers, and a standard referee would have reported dramatic reward hacking that did not occur. Studies of reward hacking should validate their referee on the outputs of the trained model.

Reproducibility and disclosure

All code, configurations, saved outputs, audits, reachability reports, the re-scoring script and the pre-registration with dated amendments are available at https://github.com/Himanshu-Sangshetti/format-drift-rl. Re-scoring the released outputs and redrawing every figure requires only a CPU. The reported runs used about $38 of cloud GPU time (Modal, NVIDIA L40S); rehearsals on free Kaggle GPUs are not reported. The study was designed, run and analysed with substantial assistance from an AI assistant (Claude, Anthropic), including code, analysis and drafting. The author reviewed all of it and takes responsibility for the content.

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A Pre-registration amendments

All amendments are dated and committed in PREREG.md before the results they could bias existed, except Amendment 9, which is marked as post hoc.

1. The gold label requires a single boxed answer; reachability becomes the primary selection measure; a density-matched control is added.
2. Qwen3-4B replaces Qwen3-1.7B as the judge after calibration (the smaller judge accepted 96% of wrong boxed answers); the density rate is fixed from base samples.
3. The learning rate is chosen on the honest control by a smoothed collapse rule (revised before the calibration results were read); 10^{-4} is selected.
4. The causal comparison becomes the sole confirmatory test, because the cross-condition tests had too few conditions to reach significance.
5. The gold label accepts an unboxed final number, after a pilot counted unboxed correct answers as hacks; a budget-driven cut order is fixed.
6. A single model; 1,536-token answers; the MBPP environment; resumable runs; one GPU type for all reported runs.
7. Run list and cut order derived from measured cost, with a floor of four seeds per causal arm.
8. The locked run set: 19 runs, then 20 at the author's request, on L40S GPUs for 120 steps.
9. **Post hoc.** The final referee (last box holding a number, formatting removed); exploitation computed from saved steps; reachability recomputed.

B Training and evaluation settings

Setting	Value
Policy	Qwen3-0.6B, thinking mode off
Algorithm	Dr. GRPO (TRL 1.17.1), one update per batch, no KL ($\beta = 0$), unscaled advantages
Batch	8 problems × 8 samples per step; 120 steps
Sampling	temperature 1.0, top-p 1.0, top-k disabled; up to 1,536 new tokens
Optimiser	AdamW, constant learning rate 10^{-4}, no warm-up
Adapter	LoRA, rank 16, $\alpha = 32$, all linear layers
Judge	Qwen3-4B, $P(\text{YES})$ from one forward pass, reference shown
Base audits	2,048 maths answers (256 problems × 8), 1,200 code answers (150 tasks × 8)
Reachability	64 samples per exploited problem from the starting model
Hardware	one NVIDIA L40S per run; about 25 s per step; about $38 in total

C Per-run results

Run	E (robust)	E (box-matching)	Accuracy, start → end	Flagged	Correct among flagged
strict s0	+0.00	+0.00	0.70 → 0.71	0	0
substring_tail50 s0	+0.02	+0.02	0.76 → 0.57	21	1
substring_tail50 s1	+0.00	+0.07	0.61 → 0.71	30	25
substring_tail50 s2	+0.02	+0.02	0.68 → 0.73	20	9
substring_tail50 s3	−0.04	−0.04	0.54 → 0.77	40	17
tail50_slope s0	+0.00	+0.00	0.73 → 0.73	9	1
tail50_slope s1	+0.00	+0.43	0.62 → 0.77	307	294
tail50_slope s2	+0.01	+0.02	0.74 → 0.64	21	10
tail50_slope s3	−0.04	−0.01	0.61 → 0.74	23	13
tail50_density s0	−0.01	−0.01	0.67 → 0.73	11	4
tail50_density s1	+0.00	−0.00	0.68 → 0.75	33	30
tail50_density s2	+0.03	+0.02	0.71 → 0.69	23	3
tail50_density s3	−0.02	−0.02	0.55 → 0.78	14	3
judge_ref_prob s0	+0.03	+0.53	0.73 → 0.71	208	190
substring_all s0	+0.42	+0.45	0.73 → 0.43	320	33
substring_all s1	+0.07	+0.59	0.62 → 0.64	512	397
mbpp_plus s0	+0.00	+0.00	0.25 → 0.57	0	0
mbpp_base s0	+0.06	+0.06	0.27 → 0.49	90	0
mbpp_base s1	+0.05	+0.05	0.23 → 0.55	84	0
mbpp_base s2	+0.01	+0.01	0.22 → 0.44	110	0

Table 6. All 20 runs. "Flagged" counts paid answers (all saved steps) marked wrong by the box-matching referee; the code referee did not change, so code runs have no false alarms.

D Referee test cases

The robust referee is tested on 18 cases, including the following (gold in parentheses; expected label after the arrow): $\boxed{\boxed{\mathbf{110}}}$ (110) → right; $\boxed{60\ \text{ square units}}$ (60) → right; $\boxed{y = 28}$ (28) → right; $\boxed{973}\ \boxed{\text{as an integer}}$ (973) → right; $\boxed{\boxed{\frac{50}{5}} = \boxed{10}}$ (10) → right; so $\boxed{21}$. 12 (12) → wrong; $\boxed{12}\ \boxed{13}$ (12) → wrong; cool dads = 12 ... $\boxed{21}$ (12) → wrong; $\boxed{2\sqrt{3}}$ (12) → wrong; a reply with no number (12) → wrong.